

NAMANGAN, Uzbekistan

WMO: 386110

Lat: 40.983N

Lon: 71.562E

Elev: 1555

StdP: 13.89

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.6	21.0	10.2	9.8	20.0	14.3	12.0	23.5	15.3	36.1	11.2	35.7	3.2	340	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	98.9	69.4	96.7	68.9	94.7	68.4	73.3	91.6	71.8	91.0	70.4	89.9	5.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	105.7	85.0	65.0	98.0	83.3	63.2	91.9	82.1	38.1	91.7	36.6	91.1	35.4	90.0	91.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	12.1	9.6	13.6	103.3	6.2	1.9	9.2	104.6	5.6	105.7	2.1	106.8	-2.4	108.1
			12.5	77.1	6.0	3.9	8.2	79.9	4.7	82.2	1.3	84.4	-3.1	87.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	59.9	33.4	38.4	51.6	62.6	72.3	80.5	83.4	80.9	72.1	59.9	46.5	35.6	(6)
(7)		DBStd	18.77	6.65	8.10	7.86	7.59	5.94	4.93	4.08	4.44	5.47	7.09	7.43	6.05	(7)
(8)		HDD50	1534	514	328	76	7	0	0	0	0	0	11	150	448	(8)
(9)		HDD65	3943	979	744	416	128	12	0	0	0	11	185	555	913	(9)
(10)		CDD50	5146	0	5	127	386	691	915	1037	958	663	319	45	0	(10)
(11)		CDD65	2080	0	0	2	57	238	465	572	494	224	28	0	0	(11)
(12)		CDH74	23436	0	0	22	487	2324	5608	7272	5548	1955	218	2	0	(12)
(13)		CDH80	11624	0	0	1	124	920	2962	4013	2844	723	37	0	0	(13)
(14)	Wind	WSAvg	4.7	3.4	4.1	4.9	5.4	5.5	5.7	5.2	4.9	4.8	4.5	3.9	3.4	(14)
(15)	Precipitation	PrecAvg	8.0	0.7	1.0	1.1	0.8	0.9	0.4	0.2	0.2	0.1	0.6	0.9	1.1	(15)
(16)		PrecMax	14.3	2.5	2.9	2.6	3.2	3.7	1.7	0.5	0.6	0.4	2.0	3.5	3.1	(16)
(17)		PrecMin	3.9	0.0	0.1	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	(17)
(18)		PrecStd	2.4	0.5	0.7	0.6	0.6	0.8	0.4	0.2	0.1	0.1	0.5	0.9	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	62.7	77.1	88.1	94.9	101.9	102.6	99.7	92.6	84.3	71.9	53.9	(19)
(20)		MCWB	43.7	48.8	56.4	62.4	64.9	68.6	70.2	70.6	67.7	62.7	55.9	45.5	(20)	
(21)		2%	DB	48.0	56.8	72.0	83.8	91.6	98.4	100.0	96.7	89.8	79.3	65.6	49.8	(21)
(22)		MCWB	41.8	46.1	54.7	61.9	64.5	67.9	69.8	69.8	65.8	60.6	52.9	43.1	(22)	
(23)		5%	DB	44.9	52.9	68.0	79.7	88.6	95.4	97.2	94.8	87.5	75.7	61.0	46.7	(23)
(24)		MCWB	39.7	44.1	53.4	60.2	63.9	67.3	69.4	69.5	64.8	58.1	51.3	41.2	(24)	
(25)		10%	DB	42.2	49.4	64.2	75.6	85.6	93.0	95.1	92.4	84.4	72.1	57.4	44.2	(25)
(26)		MCWB	38.2	42.5	51.9	58.8	63.1	66.7	68.8	68.8	63.9	56.6	49.2	39.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.9	49.8	58.5	66.8	68.7	73.5	78.4	75.2	70.5	64.4	57.4	47.4	(27)
(28)		MADB	50.2	60.6	71.3	81.8	88.4	94.1	94.9	92.6	87.5	79.3	68.4	52.0	(28)	
(29)		2%	WB	42.3	47.2	56.4	64.2	67.0	71.1	73.9	73.0	68.2	61.4	54.3	44.1	(29)
(30)		MADB	46.5	54.6	68.5	78.1	85.9	91.6	92.3	90.5	85.4	76.1	62.7	47.9	(30)	
(31)		5%	WB	40.4	45.1	54.7	62.3	65.7	69.7	71.9	71.7	66.2	59.7	52.1	41.9	(31)
(32)		MADB	43.8	51.4	65.6	75.3	83.7	90.7	91.9	90.1	83.8	73.2	59.5	45.8	(32)	
(33)		10%	WB	38.5	43.1	52.9	60.4	64.5	68.3	70.4	70.3	64.6	57.9	50.1	40.0	(33)
(34)		MADB	41.8	48.5	62.7	72.7	81.2	88.4	90.6	88.8	81.8	69.5	56.6	43.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	11.3	13.4	16.6	18.9	21.3	22.9	22.3	22.0	22.3	19.9	15.2	11.1	(35)	
(36)		5% DB	MCDBR	14.5	18.5	22.5	24.3	25.5	25.5	24.4	23.7	24.3	23.4	20.3	15.5	(36)
(37)		MCWBR	10.1	11.2	11.5	10.7	9.4	9.2	9.8	10.2	10.9	11.0	11.5	10.6	(37)	
(38)		5% WB	MCDBR	12.7	16.4	19.7	20.9	22.5	23.2	22.8	22.6	23.1	20.9	17.5	13.7	(38)
(39)		MCWBR	9.4	10.5	10.6	9.9	9.2	9.6	10.4	10.5	11.1	10.3	10.5	9.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.361	0.412	0.488	0.508	0.494	0.493	0.476	0.446	0.437	0.478	0.434	0.363	(40)	
(41)		taud	2.179	2.047	1.883	1.870	1.962	1.979	2.023	2.102	2.060	1.898	1.977	2.153	(41)	
(42)		Ebn at Noon	241	244	238	244	252	252	255	259	251	217	207	227	(42)	
(43)		Edh at Noon	32	43	57	62	57	56	54	48	47	49	40	31	(43)	
(44)	All-Sky Solar Radiation	RadAvg	643	911	1244	1664	2023	2310	2355	2135	1740	1128	713	547	(44)	
(45)		RadStd	58	101	106	141	128	101	84	96	64	118	94	70	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.63	N/A	N/A	+1.12	-1.17	-1.82	N/A	N/A	+99	+88	(46)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	+0.92	N/A	N/A	N/A	N/A	+173	+115	(47)

Nomenclature: See separate page